

IT436 – Cloud Computing Networking

Cloud Computing *- FCoE*

Acknowledgment: The presentations contains some figures and text from the following:
Cloud Computing Theory and Practice book by Dan C. Marinescu
Information Storage and Management, John Wiley & Sons, Inc.
Cloud Networking Understanding Cloud-based Data Center Networks by Gary Lee

What is FCoE?

- A SAN that transports FC data along with regular Ethernet traffic over a Converged Enhanced Ethernet (CEE) network
- Uses FCoE protocol that encapsulates FC frames into Ethernet frames
- Ensures lossless transmission of FC traffic over Ethernet

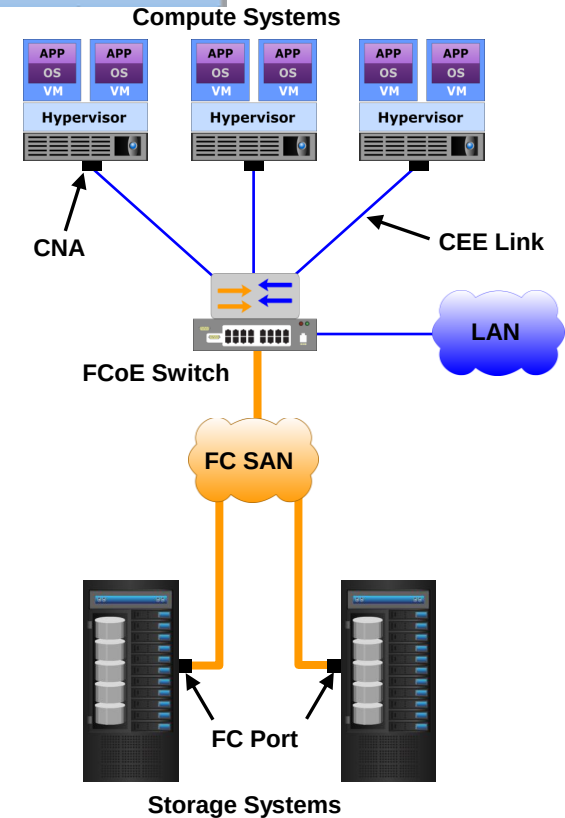
Drivers for FCoE SAN

Multi-function network components are used to transfer both compute-to-compute and FC storage traffic

- Reduces the complexity of managing multiple discrete networks
- Reduces the number of network adapters, cables, and switches required in a data center
- Reduces power and space consumption in a data center

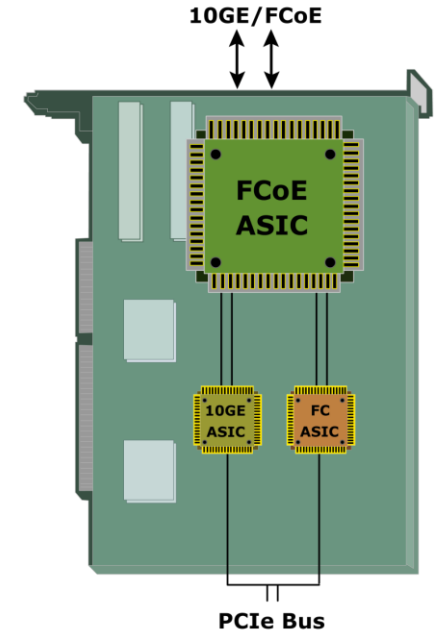
Components of FCoE SAN

- Network adapters
 - ✓ Example: Converged Network Adapter (CNA) and software FCoE adapter
- Cables
 - ✓ Example: Copper cables and fiber optical cables
- FCoE switch



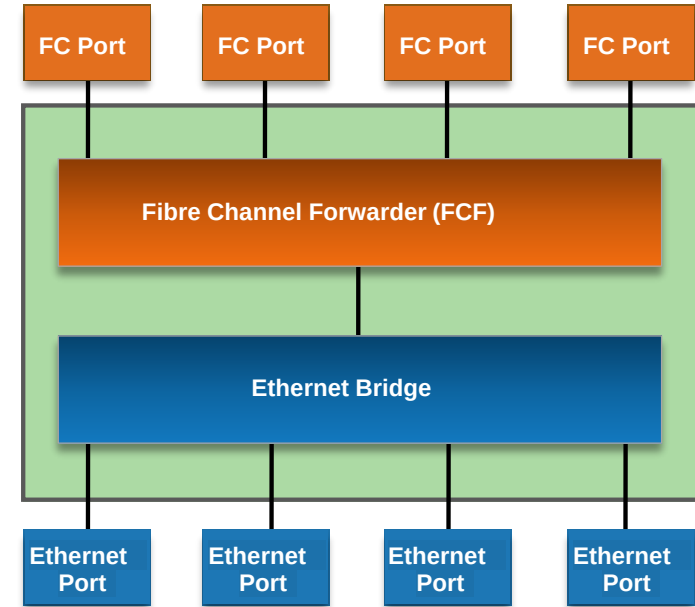
CNA and Software FCoE Adapter

- CNA:
 - ✓ A physical adapter that provides functionality of both NIC and FC HBA
 - ✓ Encapsulates FC frames into Ethernet frames and forwards them over CEE links
 - ✓ Contains separate modules for 10 GE, FC, and FCoE ASICs
- Software FCoE adapter:
 - ✓ An OS/hypervisor-resident software that performs FCoE processing
 - ✓ Supported NICs transfer both FCoE traffic and regular Ethernet traffic



FCoE Switch

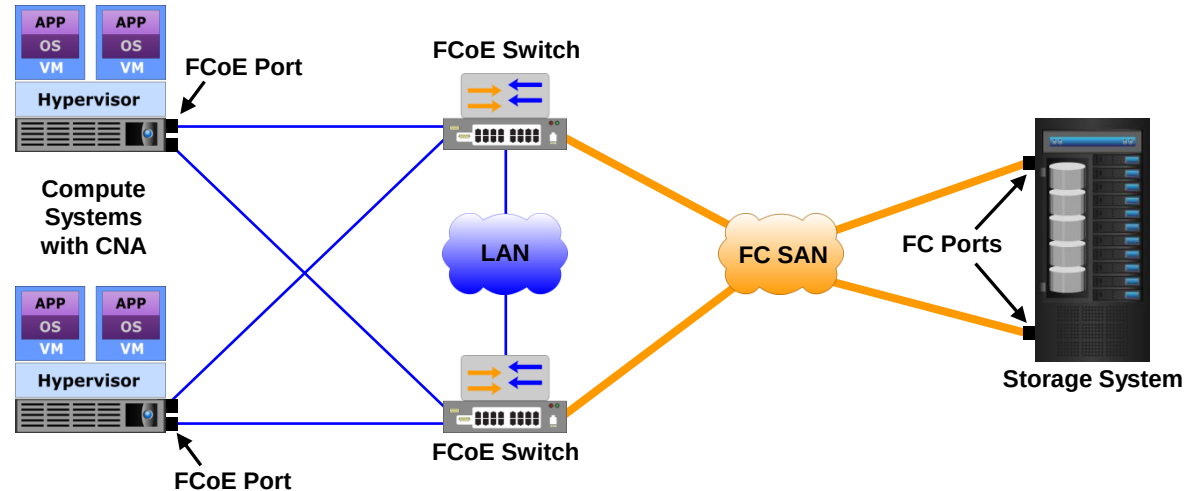
- Provides both Ethernet switch and FC switch functionalities
- Consists of FCF, Ethernet bridge, and ports for FC and Ethernet connectivity
 - FCF functions as the communication bridge between CEE and FC networks
 - ✓ Handles FCoE login requests, applies zoning, and provides fabric services
 - ✓ Encapsulates and decapsulates FC frames
- Forwards frames based on Ethertype



FCoE SAN Connectivity

FCoE with Existing FC SAN

- FCoE switches interconnect a CEE network containing compute systems with an FC SAN containing storage systems
- Suitable for environment with existing FC SAN

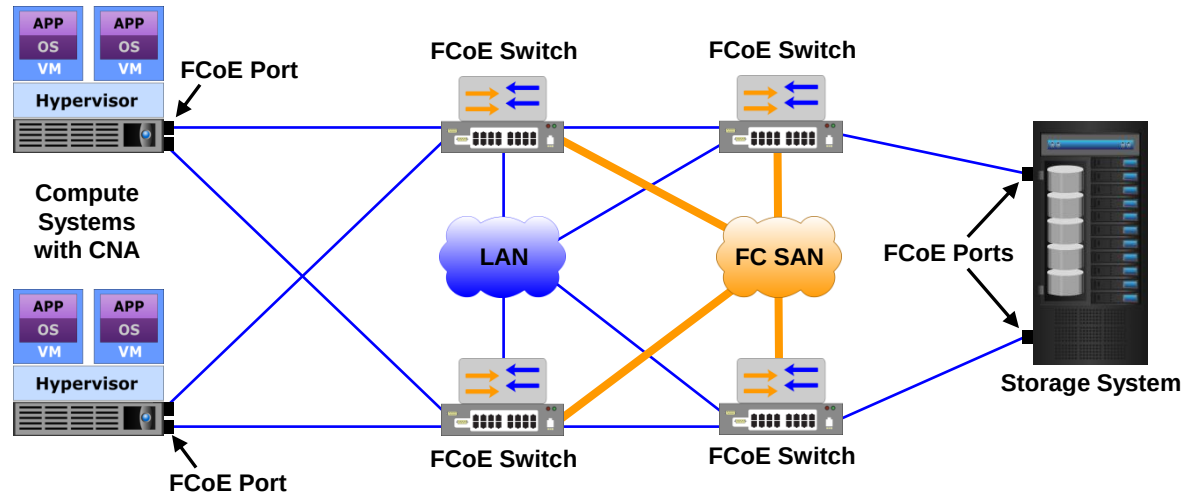


FCoE SAN Connectivity

End-to-end FCoE

FCoE switches interconnect FCoE compute systems with FCoE storage systems

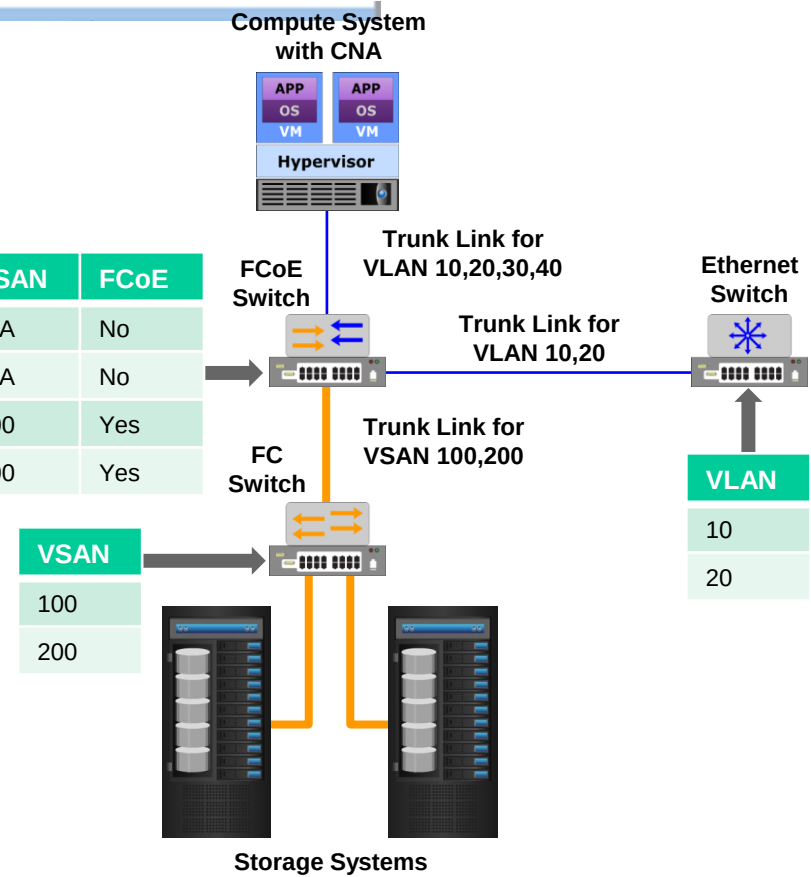
Suitable for new FCoE deployments



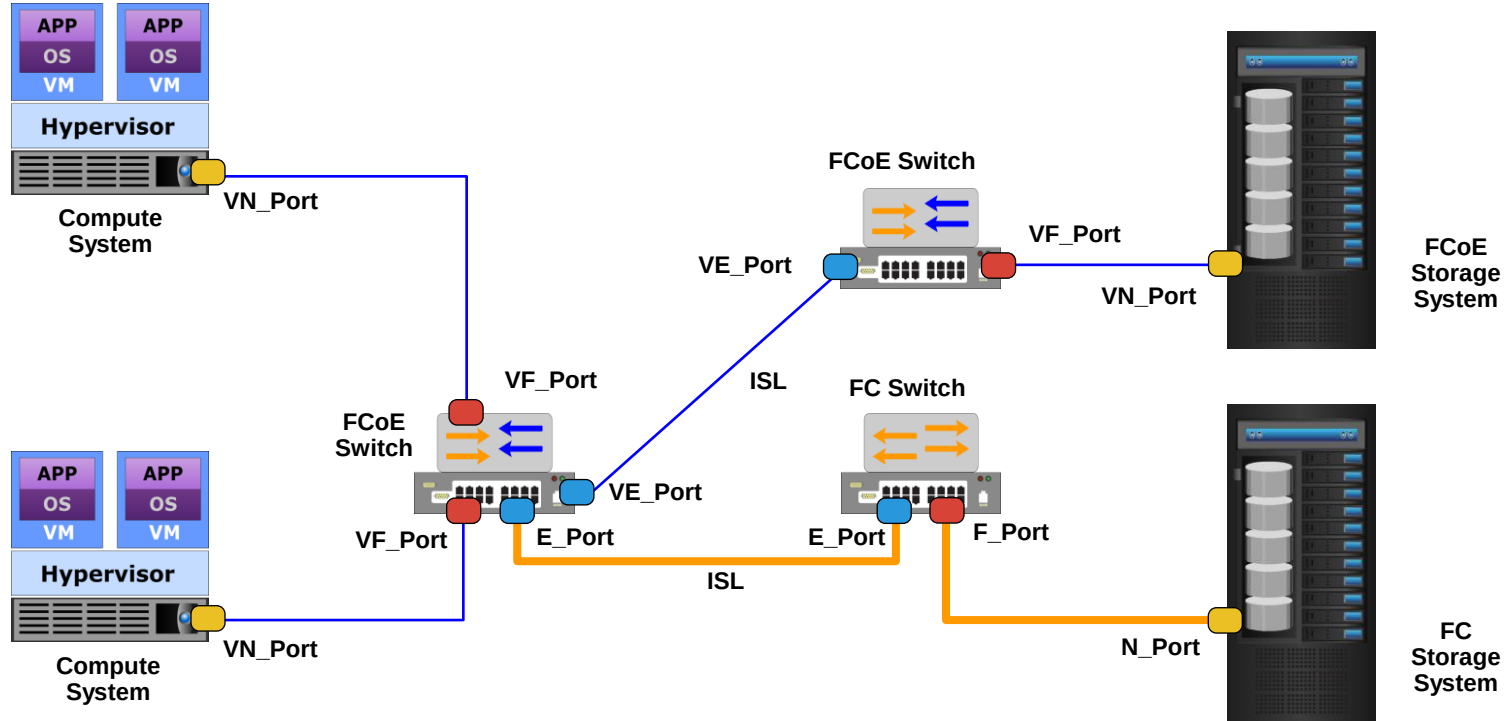
VLAN and VSAN in FCoE

- Both VLAN and VSAN may exist in the FCoE with existing FC SAN environment
- FCoE switch supports VSAN to VLAN mapping
- A dedicated VLAN configured for each VSAN
- VLANs configured for VSANs should not carry regular LAN traffic

VLAN	VSAN	FCoE
10	N/A	No
20	N/A	No
30	100	Yes
40	200	Yes



Port Types in FCoE

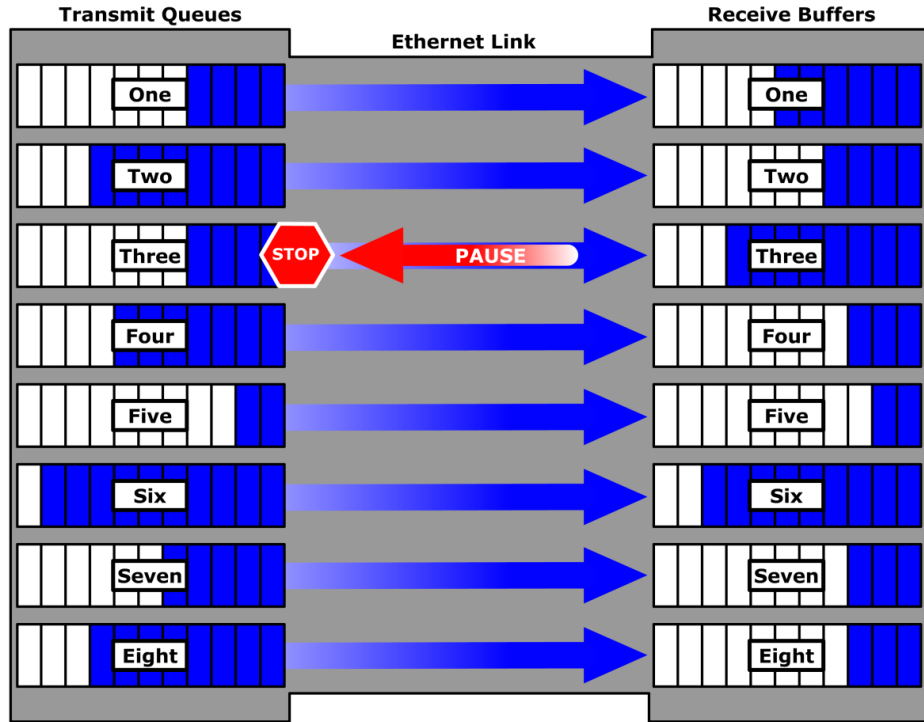


Converged Enhanced Ethernet (CEE)

- CEE refers to a set of enhancements to conventional Ethernet for making it lossless and for converging various traffic types
- CEE ensures lossless transmission of FC data over an Ethernet network
 - ✓ Eliminates the dropping of frames due to congestion
- CEE requires these functionalities:
 - ✓ Priority-based flow control (PFC)
 - ✓ Enhanced transmission selection (ETS)
 - ✓ Congestion notification (CN)
 - ✓ Data center bridging exchange protocol (DCBX)

Priority-based Flow Control (PFC)

- Creates eight virtual links on a single physical link
- Uses PAUSE capability of Ethernet for each virtual link
- A virtual link can be paused and restarted independently
- PAUSE mechanism is based on user priorities or classes of service

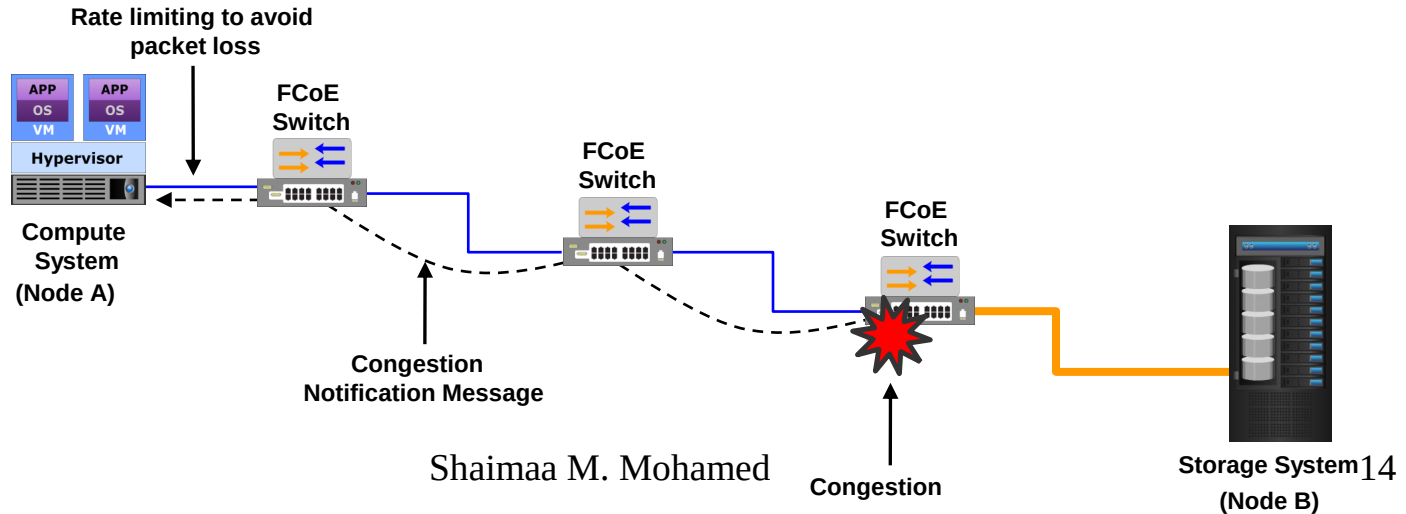


Enhanced Transmission Selection (ETS)

- Allocates bandwidth to different traffic classes such as LAN, SAN, and Inter Process Communication (IPC).
- Provides available bandwidth to other classes of traffic when a particular class of traffic does not use its allocated bandwidth

Congestion Notification (CN)

- Provides a mechanism for detecting network congestion and notifying the source
 - ✓ Enables a switch to send a signal to other ports that need to stop or slow down their transmissions

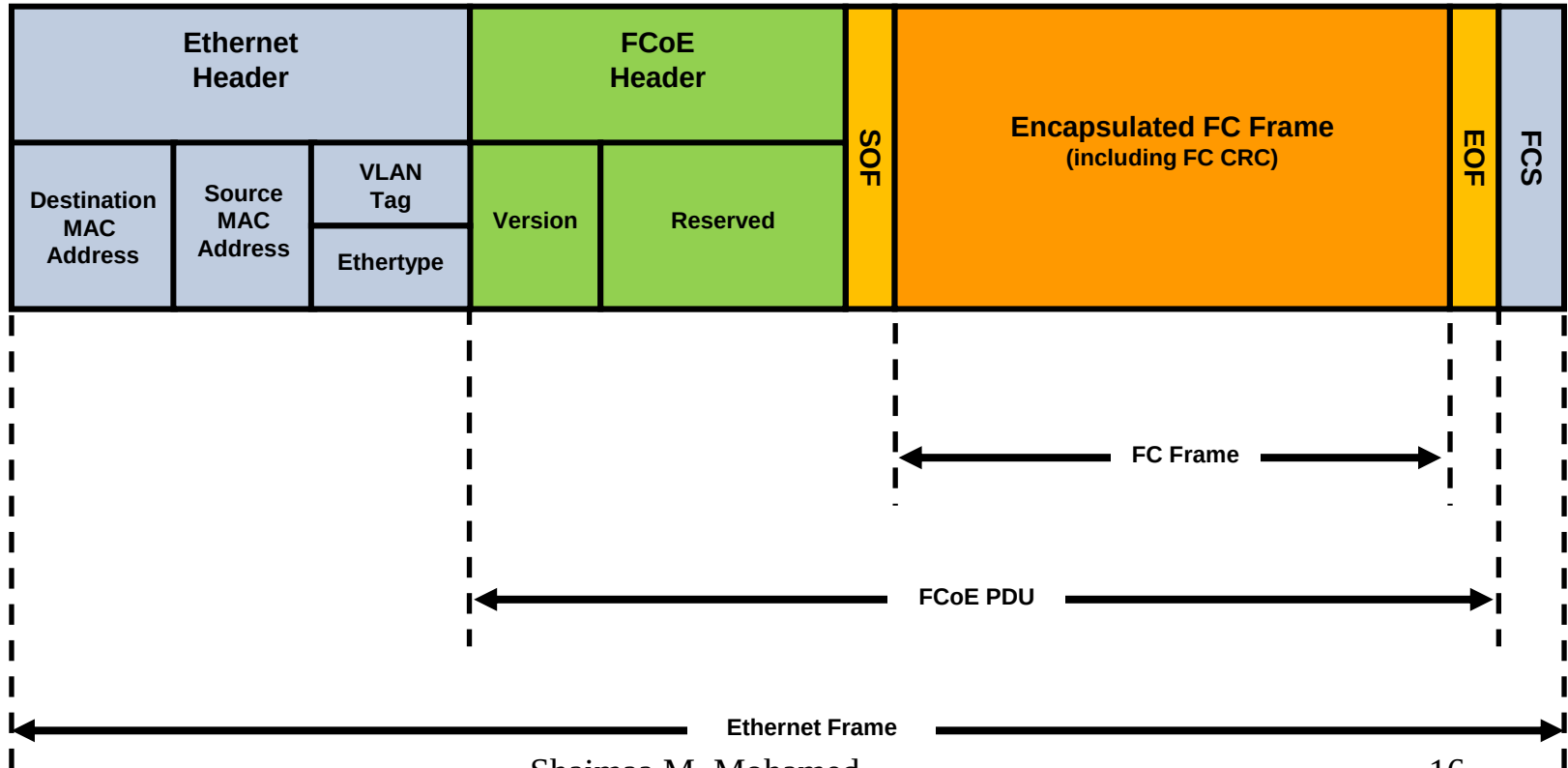


Data Center Bridging Exchange Protocol (DCBX)

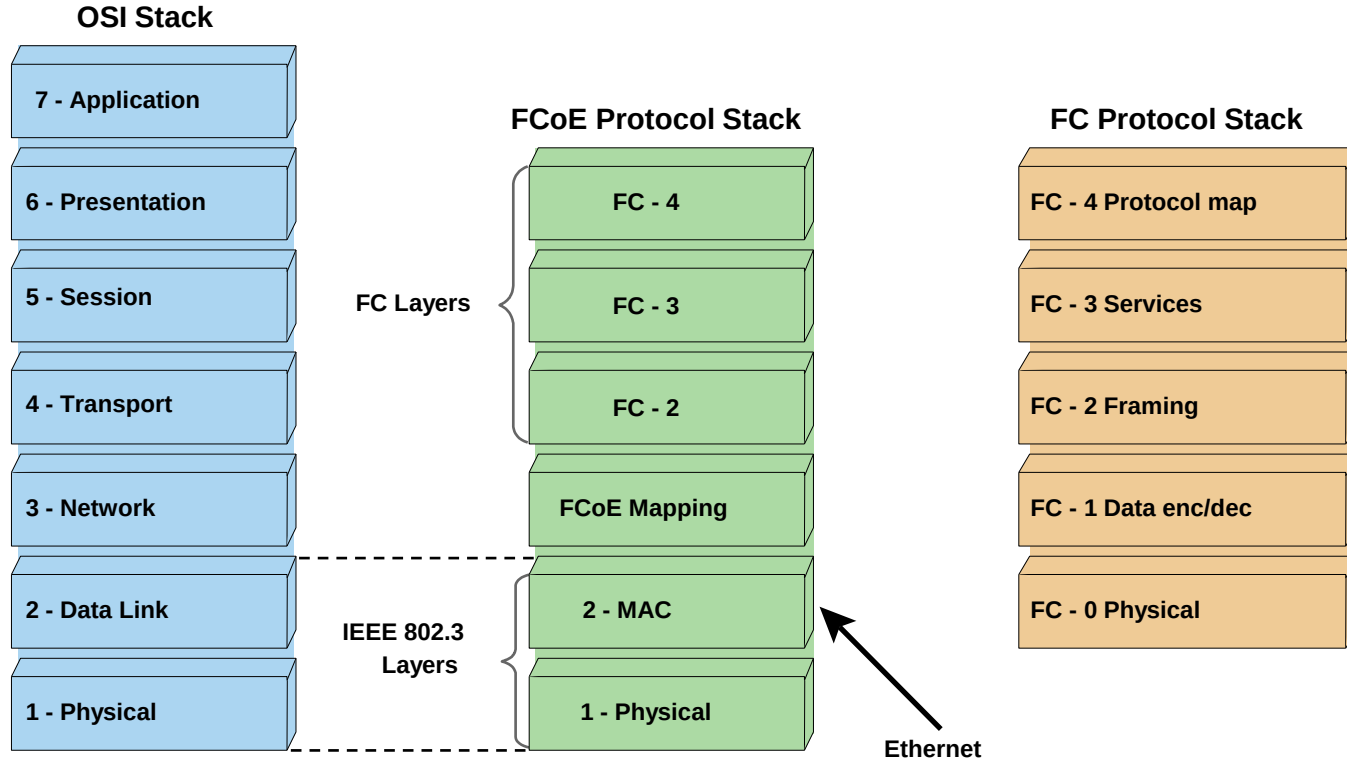
- Enables CEE devices to convey and configure their features with other CEE devices in the network
 - ✓ Allows a switch to distribute configuration values to attached network adapters
- Ensures consistent configuration across a network

FCoE Architecture

FCoE Frame Structure



FCoE Frame Mapping



FCoE Process

- FCoE process includes three key phases:
 - Discovery phase:
 - ✓ FCFs discover each other and form an FCoE fabric
 - ✓ FCoE nodes find available FCFs for login
 - ✓ FCoE nodes and FCFs discover potential VN_Port to VF_Port pairing
 - Login phase:
 - ✓ Virtual FC links are created – VN_Port to VF_Port, VE_Port to VE_Port
 - ✓ VN_Port performs FC login and obtains FC address
 - ✓ VN_Port obtains a unique MAC address
 - Data transfer phase:
 - ✓ VN_Ports transfer regular FC frames (encapsulated) over CEE network

FCoE Initialization Protocol (FIP)

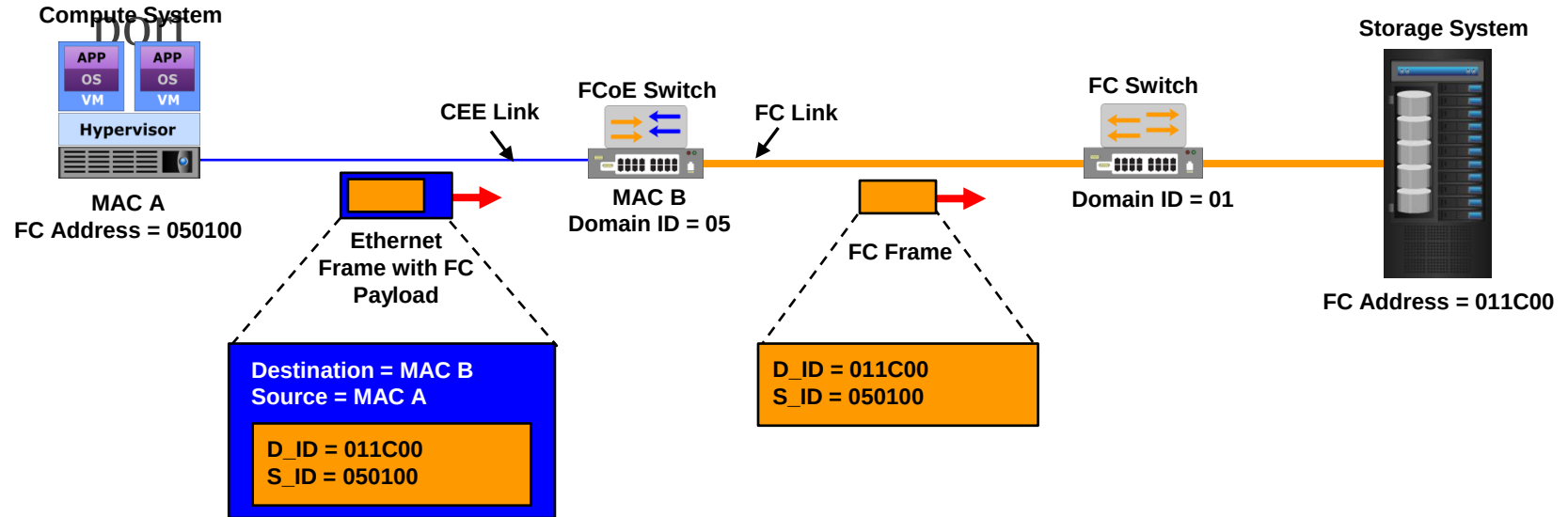
- FIP is used for discovering FCFs and establishing virtual links between FCoE nodes and FCoE switches
- FIP frames do not transport FC data, but contains discovery and login parameters
- Discovery and login use these FIP operations:
 - ✓ FCoE node sends multicast FIP Solicitation frame to find FCFs for login
 - ✓ Each FCF replies to FCoE node by sending unicast FIP Advertisement frame
 - ✓ FCoE node sends FIP FLOGI request to an appropriate FCF
 - ✓ FCF sends FIP FLOGI Accept containing FC and MAC addresses for VN_Port

FCoE Addressing

- An FCoE SAN uses MAC address for frame forwarding
 - ✓ MAC addresses are assigned to VN_Ports, VF_Ports, and VE_Ports
- VF_Ports and VE_Ports obtain MAC addresses from FCoE switch
- FCoE supports two types of addressing for VN_Ports:
 - ✓ Server-provided MAC address (SPMA)
 - Compute systems provide MAC addresses to associated VN_Ports as per Ethernet standards
 - MAC addresses are either burned-in or are configured by administrators
 - ✓ Fabric-provided MAC address (FPMA)
 - FCoE switches provide MAC addresses to VN_Ports dynamically during node login
 - MAC address has two parts: 24-bit FC-MAP and 24-bit FC address

FCoE Frame Forwarding

- To forward frame, a node must know both the MAC address of FCoE switch port and the FC address of destination node





Questions?